

BANKNOTE AUTHENTICATION

Machine Learning Project Report

ML II – Unsupervised Learning Capstone

Contributors

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Introduction & Dataset

Problem Statement

Currency counterfeiting undermines monetary systems and disproportionately harms cash-dependent communities. This project investigates whether unsupervised machine learning can reliably distinguish authentic banknotes from counterfeits using digitally extracted image features, without access to ground-truth labels during training.

Dataset

The UCI Banknote Authentication dataset comprises wavelet-transformed features extracted from scanned banknote images. After removing 24 duplicate rows, the working dataset contains

- 1,348 observations across four continuous features: variance, skewness, kurtosis, and entropy of the Wavelet Transformed image.
- One binary ground-truth label (0 = Authentic: 738 samples; 1 = Forged: 610 samples), withheld from all clustering steps.
- Class balance of 54.7 % authentic vs. 45.3 % forged — mild imbalance, favourable for centroid-based methods.
- No missing values detected; all features are numeric and on different scales, necessitating standardisation.

The wavelet transform is a signal-processing technique that captures both frequency and spatial information simultaneously, making it more discriminative than raw pixel intensities for counterfeit detection.

Algorithms Overview

K-Means Clustering: A centroid-based partitioning algorithm that assigns each point to its nearest cluster centre and iteratively refines centroids. Applied with $k = 2$ (matching the known binary structure), $\text{random_state} = 42$, and $n_init = 10$.

DBSCAN: Density-Based Spatial Clustering of Applications with Noise. Groups points in high-density regions and labels low-density points as noise. Requires no prior specification of k and naturally handles non-convex cluster shapes.

PCA: Principal Component Analysis was applied post-standardisation to project the four-feature space into two principal components, retaining 79.3 % of total variance. PCA enables 2D visualisation and was evaluated as a preprocessing variant for both algorithms.

Principal Component	Explained Variance Ratio	Cumulative
PC1	46.81 %	46.81 %
PC2	32.49 %	79.29 %

Table 1: PCA explained variance — two components retain 79.3 % of total variance.

Methodology & Pipeline

Preprocessing Steps

- **Duplicate Removal:** Deduplication:
24 identical rows removed prior to feature extraction to prevent density inflation.
- **Target Separation:** Target Separation:
The class column was excluded from all clustering inputs, preserving the unsupervised nature of the task.
- **Outlier Flagging:** Outlier Flagging (IQR):
Rather than deleting outliers — which destroys natural density structure — rows where any feature exceeded $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$ were flagged with a binary outlier_flag feature (91 rows, 6.75 %).
- **Standardisation:** StandardScaler:
All features scaled to mean = 0, std = 1 using sklearn's StandardScaler. Critical for both K-Means (Euclidean distance) and DBSCAN (epsilon neighbourhood).
- **Imputation:** Median Imputation:
Applied via sklearn Pipeline for robustness; no missing values were present in this dataset.

Hyperparameter Selection

K-Means — Elbow & Silhouette Analysis: k was set to 2 to match the binary ground truth. Silhouette scores were computed across $k \in \{2, \dots, 10\}$; k = 2 yields the highest average silhouette (0.328 on original data, 0.394 on PCA data), confirming two clusters as the natural partition.

DBSCAN — k-Distance Elbow Plot: For each sample, the distance to its 15th nearest neighbour was computed and sorted in ascending order. The elbow of this k-distance graph identifies the optimal eps — the point where distances increase sharply, indicating the boundary between core and border points:

- Original (4-feature) space: elbow at approximately 0.5 → eps = 0.5, min_samples = 15.
- PCA (2-feature) space: compressed distances shift the elbow → eps = 0.6, min_samples = 15.

Silhouette Score Objective: DBSCAN configurations producing fewer than 2 clusters were discarded. Among valid configurations, the silhouette score was used as the selection criterion, penalising runs with excessive noise.

Code Organisation

- notebooks/01_ed.ipynb — exploratory analysis, pairplots, correlation heatmap, class distribution.
- notebooks/02_preprocessing.ipynb — sklearn Pipeline, IQR flagging, StandardScaler, PCA.
- notebooks/03_kmeans.ipynb — K-Means fitting, elbow/silhouette plots, cluster visualisations.
- notebooks/04_dbscan.ipynb — k-distance graph, DBSCAN fitting, noise analysis.
- notebooks/05_evaluation.ipynb — unified metric table, confusion matrices, comparative bar charts.
- src/pipeline.py — reusable preprocessing pipeline module.
- src/metrics.py — Silhouette, Davies-Bouldin, Calinski-Harabasz, ARI, NMI wrappers.

Results & Visualisations

K-Means Evaluation Metrics

Metric	Original Data	PCA-Reduced
Silhouette Score ($\uparrow$)	0.3280	0.3941
Davies-Bouldin Index ($\downarrow$)	1.2042	1.0104
Calinski-Harabasz Index ($\uparrow$)	788.77	803.92
Adjusted Rand Index ($\uparrow$)	-0.0001	0.0011
Normalised Mutual Info ($\uparrow$)	0.0060	0.0067
Cluster Alignment Accuracy	50.96 %	51.78 %

Table 2: K-Means internal and external metrics across both feature spaces.

K-Means with PCA marginally outperforms the original-data variant on all internal metrics (Silhouette +0.066, Davies-Bouldin -0.194 , Calinski-Harabasz +15.15). However, near-zero ARI ($-0.0001 / 0.0011$) and NMI ($0.006 / 0.007$) confirm that neither configuration's clusters align with the authentic/forged split. A $\sim 51\%$ cluster alignment accuracy is indistinguishable from random chance on a balanced binary dataset.

DBSCAN Evaluation Metrics

Metric	No PCA (eps=0.5)	With PCA (eps=0.6)
Clusters Found	5	1
Noise Points	222 (16.5 %)	1 (0.1 %)
Silhouette Score	0.1957	N/A
Davies-Bouldin Index	0.9776	N/A
Calinski-Harabasz Index	257.76	N/A
Adjusted Rand Index ($\uparrow$)	0.5493	-0.0003
Normalised Mutual Info ($\uparrow$)	0.5534	0.0013

Table 3: DBSCAN metrics — PCA collapses the algorithm to a single cluster.

DBSCAN without PCA is the standout performer. ARI = 0.549 and NMI = 0.553 are the highest external scores across all four configurations, indicating meaningful cluster-to-label alignment. The 16.5 % noise rate (222 points) corresponds to ambiguous banknotes at density boundaries — consistent with the 91 IQR-flagged outliers from preprocessing plus additional borderline samples.

Visualisations on github report

- Box Plots by Class: Side-by-side box plots for each feature grouped by authentic vs. forged label, showing distributional differences and outlier patterns.
- PCA Explained Variance Bar Chart: A bar chart of PC1 (46.8%) and PC2 (32.5%) variance ratios with a cumulative line, justifying the 2-component choice
- PCA Biplot: samples in PC1-PC2 space with feature loading arrows — shows which original features drive each component.
- K-Means Cluster Plot (PCA): scatter of PC1 vs. PC2, points coloured by cluster assignment, centroids marked with $\times$.
- Confusion Matrix Heatmaps: 2×2 annotated heatmaps (seaborn.heatmap, annot=True) comparing best cluster assignment to ground truth for both algorithms.
- Grouped Bar Chart of Metrics: four metric groups, bars for each of the three valid model variants — immediately surfaces which model wins on each axis.

- Radar / Spider Chart: A radar chart with vertices for all 5 metrics normalised to [0,1], one polygon per model. Ideal for executive summary slides.

Comparative Discussion

K-Means — Strengths & Weaknesses

K-Means is computationally efficient and produces deterministic results (with fixed `random_state`). Its internal metrics (Silhouette 0.394, Davies-Bouldin 1.01, Calinski-Harabasz 803.92 on PCA data) suggest reasonably compact, well-separated clusters — but these describe geometric density, not class structure. The near-zero external metrics expose the fundamental limitation: K-Means partitions by proximity to centroids, not by the density contours that separate authentic from forged notes.

Strengths: Fast convergence, intuitive centroids, stable across runs (`n_init = 10`). Weaknesses: Assumes convex, equally-sized clusters; sensitive to initialisation; cannot detect noise or outliers; cluster labels are arbitrary and require post-hoc alignment.

DBSCAN — Strengths & Weaknesses

DBSCAN without PCA achieves the best external alignment (ARI 0.549, NMI 0.553), confirming that authentic and forged banknotes form high-density regions in the 4-feature space. The five discovered clusters suggest multi-modal structure within the two classes — consistent with multiple counterfeiting techniques or printing batches producing internally similar but cross-batch distinct notes.

Strengths: No prior `k` specification, natural noise handling, discovers non-convex shapes, outlier identification aligns with preprocessing flags. Weaknesses: Sensitive to `eps`; lower internal Silhouette (0.196) reflects that five unequal clusters score worse by centroid-distance metrics; computationally heavier on large datasets.

Metric Agreement & Disagreement

Internal metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) consistently favour K-Means with PCA, since these metrics reward compact, spherical, well-separated clusters — the geometry K-Means explicitly optimises for. External metrics (ARI, NMI) tell the opposite story: DBSCAN without PCA dominates, revealing that the compact clusters K-Means finds do not align with the supervised labels.

This disagreement is a textbook illustration of the unsupervised evaluation paradox: a model can appear excellent on internal measures while failing to recover ground-truth structure. For a security-critical task like banknote authentication, external alignment is the operationally relevant criterion.

Impact of PCA on Clustering Quality

PCA benefits K-Means by projecting data into a lower-dimensional space where variance is maximised along orthogonal axes, marginally improving all internal metrics. However, PCA catastrophically degrades DBSCAN: compressing four features into two dimensions merges the distinct density regions that DBSCAN relies on. The variance discarded (20.7 %) contains precisely the density-distinguishing information DBSCAN needs, collapsing five meaningful clusters into one.

Conclusion: PCA is beneficial for visualisation and K-Means, but contraindicated for DBSCAN on this dataset.

Conclusion & Future Work

Summary of Findings

This project demonstrates a complete unsupervised ML pipeline for banknote authentication. Of the four evaluated configurations, DBSCAN without PCA ($\text{eps} = 0.5$, $\text{min_samples} = 15$) achieves the strongest alignment with ground-truth labels ($\text{ARI} = 0.549$, $\text{NMI} = 0.553$), substantially outperforming K-Means despite producing five clusters rather than two. K-Means with PCA scores best on internal metrics but fails to recover the authentic/forged split. PCA benefits K-Means marginally while destroying DBSCAN's density structure entirely.

The preprocessing pipeline — deduplication, IQR outlier flagging, median imputation, and StandardScaler normalisation — produces a clean, reproducible dataset. The 16.5 % DBSCAN noise rate corresponds to genuine ambiguity at density boundaries, consistent with IQR-flagged outliers and real-world edge-case banknotes.

Limitations

- Sensitivity to initialisation: K-Means results vary with centroid seeds; $n_init = 10$ mitigates but does not eliminate this sensitivity.
- Runtime scalability: DBSCAN's $O(n \log n)$ complexity (with spatial indexing) is acceptable at 1,348 samples but may require approximation algorithms (HDBSCAN, ball-tree) at production scale (millions of notes per day).
- Binary outlier_flag information loss: the IQR flag collapses degree of outlierness into a single bit, reducing discriminative signal after standardisation.
- Fixed distance metric: both algorithms use Euclidean distance; wavelet-feature data may benefit from Mahalanobis or cosine distance, which account for feature correlation and scale jointly.
- No label supervision: external metrics can only be computed in retrospect; in deployment, ground truth would be unavailable for new note batches.

Suggested Improvements

1. Continuous Anomaly Scoring: Replace the binary IQR outlier_flag with a continuous Isolation Forest or Local Outlier Factor (LOF) score. This provides a smooth anomaly signal in $[-1, 0]$, improving DBSCAN density estimation and K-Means distance calculations without the arbitrary $1.5 \times \text{IQR}$ threshold.

2. Automated Hyperparameter Tuning: Grid-search DBSCAN $\text{eps} \in [0.3, 1.0]$ and $\text{min_samples} \in \{5, 10, 15, 20\}$ using Silhouette Score as the objective, restricting to runs where ≥ 2 clusters are found. Apply Elbow and Silhouette Analysis for K-Means across $k \in \{2, \dots, 10\}$, removing dependence on the known ground-truth class count.

3. Ensemble Clustering: Build a co-association matrix from multiple K-Means, DBSCAN, and Agglomerative Clustering runs with varied hyperparameters. Apply spectral clustering on the consensus matrix to derive stable final labels, combining centroid, density, and hierarchical perspectives and reducing sensitivity to individual algorithm assumptions.

4. Alternative Distance Metrics: Evaluate Mahalanobis distance (accounts for inter-feature correlation) and cosine distance (emphasises angular separation) as replacements for Euclidean distance in both K-Means and DBSCAN, particularly relevant given the moderate correlations observed in the feature correlation heatmap.

5. Supervised Production Model: Since verified labels are available, a Random Forest classifier ($n_estimators = 200$, $\text{class_weight} = \text{'balanced'}$) should be deployed for production authentication, achieving $\sim 100\%$ test accuracy with 5-fold CV accuracy of $99.70\% \pm 0.43\%$. Unsupervised methods remain valuable for anomaly detection on unlabelled incoming batches and for exploring novel counterfeiting pattern clusters.